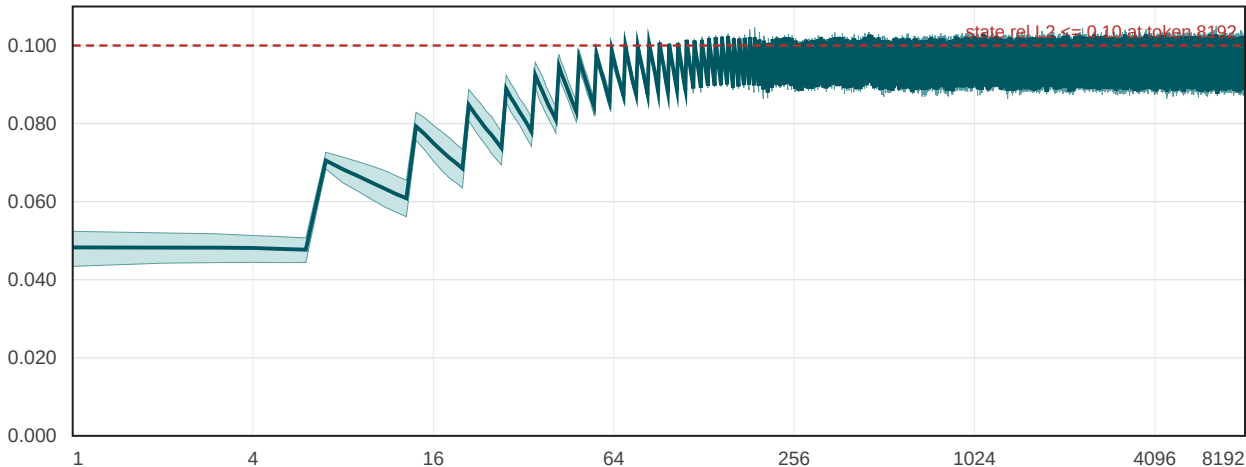


Held-out synthetic recurrent-state drift

Six-trace min-max envelope Six-trace mean Preregistered engineering gate

Selected: MXFP4 RS2 activation/base + sparse residual + MXFP8-E4M3 log, R=7

State relative L2 error versus FP32



Decode token index (log scale)

Held-out nominal synthetic Q/DQ; 3 seeds x {random,zero} S0; S is KxV; rel-L2 floor=1e-12